Friction Before Failure: Making Review Burden Visible in Regulated Document Work

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Abstract

Deviation and Corrective Action and Preventive Action (CAPA) review in pharmaceutical manufacturing environments involves multiple reviewers, iterative revision cycles, and formal approval chains — yet the coordination work that produces delay remains invisible to any single participant. This paper argues that review friction is not merely a coordination inefficiency but a structurally hidden form of work whose invisibility shapes accountability conditions before any formal quality event occurs. I report a two-part prototype-based study in which I developed an external workflow observability layer over a synthetic dataset of 100 deviation-CAPA review records and evaluated it with $N = 20$ QA/QC-experienced participants. The first component is a six-tab workflow dashboard that surfaces revision burden, comment distribution, and document journey patterns as interpretable signals. The second is a TF-IDF pattern detection layer that identifies recurring issue clusters across review cycles. Neither component modifies validated documentation systems or automates quality decisions — a constraint constitutive of GMP-regulated environments. Findings show that all 20 participants anticipated reduced accountability asymmetry from friction visibility, described role-differentiated utility across submitter and reviewer positions, and rated all components 5.0/5 on usability and adoption intent. A notable distinction emerged between the checklist — rated immediately useful with no conceptual reorientation required — and the pattern detection layer, which required

initial orientation to interpret but reached equivalent ratings once demonstrated. This dissociation between initial accessibility and final endorsement is the study's central empirical finding: it is not that the pattern detection layer is less useful, but that the problem it surfaces — structural friction accumulation — is one practitioners have not previously had the tools to name. I introduce review friction visibility as a design problem specific to regulated document environments, identify the judgment-preservation constraint as a novel design boundary for workflow support research, and argue that making hidden coordination work legible is a precondition for more equitable accountability in regulated sociotechnical organizations.

Keywords: review friction, workflow visibility, CAPA, pharmaceutical manufacturing, GMP, invisible work, judgment-preserving design, regulated environments

1 Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, deviation and corrective action (CAPA) records are among the most consequential documents a quality organization produces. Under GMP, every unplanned departure from an approved process must be formally investigated and resolved: the resulting record — a Corrective and Preventive Action (CAPA) document — traces the problem, its root cause, and the actions taken to prevent recurrence. Their review involves multiple actors — quality assurance specialists, subject matter experts, department managers — cycling through iterative rounds of comment, revision, and approval. These review processes are routinely experienced as slow. But the standard diagnosis of that slowness — too many reviewers, unclear ownership, insufficient training — misidentifies the problem. The core issue is not delay itself. It is that the coordination work producing the delay is invisible.

Review friction accumulates gradually: a wording comment triggers a revision cycle; a reviewer reopens a document closed in a previous round; a pattern of minor but recurring feedback generates disproportionate turnaround time across a document class. These patterns are real, consequential, and structural — but they remain distributed across

individual comment threads, document versions, and reviewer roles. No single actor has a complete view of them. As a result, friction becomes visible only after delays have already escalated, and accountability is assigned retroactively, without access to the coordination conditions that produced the outcome.

This paper reframes workflow delay in regulated review environments as an observability problem. I argue that review friction — revision loops, comment accumulation, uneven reviewer burden — can be externalized as interpretable signals without collapsing reviewer judgment into automated outputs. The distinction matters: in GMP environments, human interpretive authority over quality decisions is not merely a design preference but a regulatory requirement. Any system that automates review conclusions or overrides reviewer discretion is, in a compliance sense, a change to the quality system itself. The design challenge is therefore to make friction visible while leaving judgment intact.

I report a two-part prototype-based study. In the first part, I developed a six-tab workflow dashboard over a synthetic dataset of 100 deviation-CAPA review records, and evaluated it with $N = 20$ QA/QC-experienced participants. In the second part, I added a computational pattern detection layer using TF-IDF similarity clustering to surface recurring issue patterns across review cycles. Neither component modifies validated documentation systems; both operate as external observability layers. Findings show that participants could identify bottlenecks earlier, that revision loops became visible as patterns rather than isolated events, and that the friction-visibility concept — though initially unfamiliar for the pattern detection component — was consistently endorsed as useful once demonstrated, with all 20 participants indicating adoption intent across both components.

The contributions of this work are threefold. First, I introduce review friction visibility as a design problem specific to regulated document environments, identifying the judgment-preservation constraint as a novel design boundary for workflow support research. Second, I demonstrate that a lightweight observability layer — combining rule-based classification with computational pattern detection — can surface hidden coordination work without automating quality decisions. Third, I offer a theoretical framing that connects HCI

concerns about workflow visibility to broader questions about how accountability is structured in regulated sociotechnical organizations, extending the argument that coordination work shapes responsibility conditions before formal judgment begins.

2 Related Work

Awareness and Coordination in Collaborative Work Systems

A foundational concern in CSCW research is how participants in distributed collaborative work maintain awareness of each other's activities. Dourish & Bellotti (1992) establish that awareness — understanding of the activities of others that provides context for one's own action — is critical to successful collaboration and is poorly served by active, explicit information-sharing mechanisms that penalize information providers and presuppose relevance. Their study of shared editing systems demonstrates that passively available awareness information, embedded in the shared workspace itself, allows participants to coordinate dynamically without the overhead of explicit communication. Applied to regulated document review, this framework implies that the coordination work occurring across review cycles — who has commented, which rounds are open, where revision burden is accumulating — should ideally be available as ambient, interpretable signal rather than requiring active compilation by individual reviewers. The prototype developed in this study is, in these terms, a passive awareness mechanism applied to an asynchronous regulated workflow.

Invisible Work and the Limits of Formal Indicators

Coordination work in distributed systems often becomes invisible precisely because it is effective: when practitioners manage revision loops, absorb comment burden, and route documents through informal channels, the effort expended does not appear in any formal record. Star & Strauss (1999) theorize this as invisible work — labor that is performed but not captured by organizational indicators, and therefore not attributed to the structural conditions that produce it. In regulated review environments, this invisibility has specific

implications: the coordination work that generates delay sits outside CAPA records, audit trails, and quality metrics. It is not measured because it precedes the formal quality events that the measurement system is designed to capture. The observability layer developed in this study makes this invisible work legible — not by automating its interpretation, but by surfacing the patterns that indicate its presence.

Situated Action and the Conditions of Review Work

The theoretical basis for treating visibility as consequential for coordination comes from Suchman's (1987) account of situated action. Suchman argues that action is not plan-driven but shaped by the immediate conditions of its execution — what is observable, reachable, and interpretable at the moment a decision must be made. Applied to document review, this implies that the quality of review decisions is shaped not only by reviewer expertise but by the conditions of visibility under which those decisions are made. A reviewer who cannot see how a document has moved through prior rounds, what patterns of comment have accumulated, or where the current cycle sits relative to historical revision burden is making decisions in conditions of reduced observability. The observability layer addresses this not by supplying the reviewer with recommendations, but by improving the conditions under which their own judgment operates.

Human-AI Interaction and Judgment-Preserving Design

The pattern detection layer in this study uses TF-IDF similarity clustering to surface recurring issue patterns across review cycles. The design of this layer is governed by a principle that Amershi et al. (2019) formalize across eighteen guidelines for human-AI interaction: AI systems should make clear what they can and cannot do, display contextually relevant information, and support human control rather than supplant it. In regulated manufacturing environments, this principle is not merely a design preference but a compliance requirement: quality decisions must be made and documented by qualified human reviewers, and any system that automates or pre-empts those decisions introduces a regulatory risk. The pattern detection layer therefore surfaces clusters and burden

scores as observability tools rather than decision outputs — making patterns interpretable without making them actionable in the automated sense.

Prototype-Based Evaluation

The methodological approach in this study builds on established HCI practice for evaluating prototype systems. Nielsen (1994) demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability problems and can be combined with post-task assessment to understand both navigational behavior and perceived utility; subsequent work confirms that $N = 20$ expert participants in domain-specific populations reliably surfaces dominant usability and friction patterns (Nielsen, 1994). Hutchinson et al. (2003) establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe — a technology probe makes visible the conditions of use it is designed to address, rather than simply testing whether a given interface is usable. The pilot evaluation in this study applies this dual function to regulated review workflows: the dashboard and pattern detection layer are evaluated not only as interfaces but as instruments for making review friction visible and discussable.

Prior work has not examined review friction visibility as a design problem in regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. This study addresses that gap directly.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation and CAPA review workflows in these environments involve multiple reviewers, iterative revision cycles, and formal approval chains — all of which generate coordination work that is rarely visible as a unified pattern to any individual participant. The study examines whether an external observability layer can make this coordination work legible without modifying validated documentation

systems or automating quality decisions.

Dataset Construction

The study used a synthetic dataset of 100 deviation-CAPA review records generated programmatically in Python. Because actual facility records are subject to non-disclosure obligations, the dataset was constructed to reflect the structural and terminological characteristics of real GMP review workflows — including comment categories, review round sequences, severity classifications, turnaround times, document reopening events, and reviewer role assignments typical of pharmaceutical quality operations — without reproducing proprietary content.

This approach preserves the ecological validity of the coordination problem under study: the synthetic records exhibit the naming conventions, comment distribution patterns, and revision cycle structures characteristic of real deviation-CAPA workflows, including the multi-round back-and-forth and uneven reviewer burden that constitute the friction patterns this study aims to surface. The judgment-preservation constraint that defines the core design boundary of this work applies equally to synthetic and live records: in GMP environments, any system that modifies, annotates, or acts on quality records must itself be validated, meaning that observability tools must operate externally over data rather than within the document management system.

Phase 1: Workflow Dashboard Prototype

Based on the synthetic dataset, I developed a six-tab workflow dashboard as a Streamlit web application. The dashboard was designed as an external observability layer — it reads from the dataset but does not modify, version, or interact with any validated documentation system. The six tabs surface distinct dimensions of review burden:

- **Overview**: comment category distribution, severity breakdown, and average turnaround time by reviewer role

- **Heatmap**: revision bottleneck visualization across reviewer-document type combinations, with loop frequency indicators
- **Checklist**: a pre-review scaffold with a rule-based comment classifier to support consistent review preparation
- **Document journey**: full review lifecycle trace with round-by-round summary for individual records
- **Before/after**: comparative workflow support visualization
- **Reviewer simulation**: comment framing tool with friction estimation to support reflection on review patterns

The design of each tab was governed by a judgment-preservation principle: the dashboard surfaces patterns and distributions as interpretable signals, but does not recommend actions, assign responsibility, or produce outputs that function as quality decisions.

Phase 2: Pattern Detection Layer

In the second phase, I added a computational pattern detection layer using TF-IDF similarity clustering. The layer operates over comment text across review cycles and generates: (1) similarity-based issue clusters that group recurring comment patterns across documents and reviewers; (2) revision burden scores that quantify the coordination load associated with individual records; and (3) plain-language pattern summaries that describe a document's review state and the stage at which bottlenecks have accumulated.

As with the dashboard, the pattern detection layer is designed as an observability tool rather than a decision-support system. It surfaces structural patterns that are difficult to detect manually due to their distributed nature, but does not recommend corrective actions, assign root causes, or produce outputs intended to substitute for reviewer judgment.

Pilot Evaluation

I evaluated both components with $N = 20$ participants recruited from a pharmaceutical quality team with QA/QC review experience. This sample size is consistent with established think-aloud methodology, in which $N = 20$ expert participants in a domain-specific population reliably surfaces dominant usability and friction patterns (Nielsen, 1994). Participants included individuals with backgrounds in quality assurance, quality control, and regulated documentation review. The evaluation was conducted outside any live validated system — a necessary condition in GMP contexts, where introducing an unvalidated interface into a production quality workflow would itself constitute a compliance risk.

Participants completed task-based evaluation scenarios using both the dashboard and the pattern detection layer, and provided feedback on purpose clarity, friction visibility, checklist usability, simulation interpretability, and adoption intent. Post-task discussion probed the conceptual accessibility of the friction-visibility framing — whether participants found the idea that review burden deserves structured visibility legible and useful, independent of interface design.

Scope and Limitations

This study is exploratory and pilot in scope. The synthetic dataset, while constructed to reflect real workflow structures, has less variance than a live record corpus would exhibit. Participants were recruited from a single team, which may have inflated acceptance ratings. The study does not establish long-term adoption effects, downstream impacts on review quality, or effects on formal quality metrics such as CAPA closure time.

4 Findings

Friction Visibility as Fairness Infrastructure

The most consequential participant response was not about interface usability but about what visibility of review burden would change organizationally. All 20 participants antici-

pated reduced accountability asymmetry from the dashboard — a finding that emerged consistently across both submitter and reviewer roles without prompting. Participants described a recurring condition in current practice: review delays accumulate, documents circulate without clear progress signals, and when closure finally fails or slows, responsibility tends to be assigned without access to the coordination conditions that produced the outcome. One participant characterized the anticipated value of the dashboard in terms of reducing situations that feel unfair — cases where a document's delay is attributed to the person who submitted it rather than to the review burden accumulated across cycles.

This response points to a finding that goes beyond usability. In GMP environments, QA review is a formal gatekeeping function: every document must be approved by a qualified reviewer before proceeding. But the standards applied across that gatekeeping function are not always consistent, and the burden carried by different reviewers at different stages is not visible to any single participant. When delays occur, they are experienced asymmetrically — felt by submitters, invisible to quality systems, and unattributed to the coordination structure that produced them. Participants anticipated that the dashboard would make this asymmetry legible: not by resolving it automatically, but by making it discussable.

Role-Differentiated Utility

Participants described the dashboard's value differently depending on their role in the review process. From the perspective of document submitters — practitioners in QC, production, or other departments who initiate deviation or CAPA records — the primary value was visibility into where their document stood and what type of review activity was generating delay. The current absence of this information means that submitters experience delay as an opaque condition, with no basis for distinguishing between a document that is actively under review and one that has been stalled.

From the perspective of QA reviewers, the value was different: the dashboard offered a way to monitor and regulate their own review workload across documents. Participants in QA roles described the possibility of using the dashboard to identify when their

review burden was accumulating unevenly, to prioritize across open records, and to adjust their own pacing in response to visible pattern information. This suggests that the observability layer has distinct utility for different positions in the review workflow — not as a unified transparency tool, but as a set of role-differentiated instruments for making coordination conditions legible. All 20 participants expressed adoption intent, with this role-differentiated framing emerging as the dominant account of why the system would be useful.

Pattern Detection and the Limits of Initial Interpretability

The TF-IDF clustering layer received a final usability rating of 5.0/5 and was endorsed by all 20 participants as useful once demonstrated — yet it required initial orientation to interpret in a way that the dashboard's other components did not. Several participants noted that clustering outputs — similarity scores, issue cluster labels, burden scores — were not immediately self-explanatory without guidance on what the categories represented and how they had been generated. This points to a design gap distinct from interface usability: the friction-visibility concept itself was the source of interpretive difficulty, not the interface.

Participants noted that the idea of representing review burden as a structured, observable signal was unfamiliar — not because it was counterintuitive, but because existing practice does not provide a referent for it. Document review is experienced as a series of individual events, not as a pattern with measurable properties. The conceptual work required to see it as such is non-trivial and cannot be resolved through interface design alone. Crucially, this initial unfamiliarity did not produce rejection — once the framing was explained, all 20 participants endorsed the concept as genuinely useful for their work.

The Checklist as the Most Immediately Legible Feature

The pre-review checklist received 5.0/5 usability ratings across all participants, with no conceptual reorientation required. Participants described it as immediately useful and directly applicable to their current work. The contrast with the pattern detection layer

is analytically significant: both components received equivalent final ratings (5.0/5) and equivalent adoption intent (20/20), but they arrived at those ratings through different pathways. The checklist was immediately accessible because it addresses a problem practitioners already recognize — inconsistent review preparation. The pattern detection layer required initial orientation because it addresses a problem practitioners have not previously had the tools to name — structural friction accumulation.

This dissociation is the study's central empirical finding. It is not that the pattern detection layer is less useful than the checklist; it is that usefulness and immediate legibility are not the same thing. A tool can be highly useful for a problem that practitioners have not yet learned to see. The implication for deployment is practical: checklist-first adoption establishes the dashboard's credibility within existing review practice, while pattern detection outputs become more legible as practitioners develop the interpretive frame to use them.

Summary

Taken together, the findings support the central claim of this study: review friction in regulated environments is not merely coordination inefficiency but a structurally hidden form of work whose invisibility shapes accountability conditions. All 20 participants rated both components 5.0/5 and expressed adoption intent, while the pattern detection layer's initial interpretive overhead — compared to the checklist's immediate accessibility — reveals that visibility tools must address not only interface design but the conceptual work of making a new problem category legible. When friction becomes visible, it becomes possible to discuss, compare, and act on it.

5 Discussion

Visibility as a Precondition for Accountability

The central finding of this study is that review friction in regulated environments is not simply a coordination inefficiency — it is a structurally hidden form of work whose

invisibility shapes accountability conditions before any formal quality event occurs. When a CAPA record is delayed, the quality system records the delay but not the coordination work that produced it. Responsibility is assigned in the absence of the conditions that would make that assignment fair: who reviewed what, when burden accumulated, which revision cycles were produced by which patterns of comment. The observability layer developed here does not resolve this asymmetry, but it makes it visible — and visibility is a precondition for accountability that existing quality systems do not currently provide.

This extends Suchman's (1987) account of situated action into the domain of quality review work. The finding that all 20 participants anticipated reduced unfairness from the dashboard is a direct instantiation of this principle: it is not that review decisions become fairer automatically, but that the conditions under which accountability is assigned become more visible, and therefore more contestable. The dashboard does not change what happened; it changes what can be seen about what happened.

Invisible Work and the Organizational Stakes of Legibility

The role-differentiated responses documented in the findings reflect the asymmetric distribution of invisible work that Star & Strauss (1999) describe. In regulated review workflows, the coordination labor of managing revision cycles, absorbing inconsistent feedback, and routing documents through informal channels is performed primarily by document submitters — practitioners in QC, production, and other departments who have the least formal power in the review hierarchy. This labor is invisible to the quality system and unattributed to the structural conditions that produce it. QA reviewers, by contrast, carry a different form of invisible burden: the coordination work of managing their own review load across multiple open records without a shared visibility mechanism.

The dashboard addresses both forms of invisibility, but in different ways. For submitters, it makes the accumulated review burden on their documents legible — transforming opaque delay into an interpretable pattern. For QA reviewers, it provides a workload monitoring function that current systems do not support. That all 20 participants across both roles reported adoption intent suggests that the observability layer is not a tool for one side

of a coordination asymmetry, but a shared infrastructure for making that asymmetry discussable.

Judgment-Preserving Design in Compliance-Constrained Environments

The design constraint that governed both components of this study — that computational outputs must support human judgment rather than substitute for it — is not merely a normative preference. In GMP environments, quality decisions must be made and documented by qualified human reviewers; any system that automates or pre-empts those decisions introduces a compliance risk that can itself become a regulatory finding. This constraint creates a design space that prior HCI work on AI-assisted decision support has not fully addressed: how to make structural patterns legible without producing outputs that function as quality decisions.

The pattern detection layer's approach — surfacing TF-IDF clusters and burden scores as observability signals rather than recommendations — is one instantiation of the judgment-preservation principle that Amershi et al. (2019) formalize. The finding that clustering outputs required initial interpretive orientation, despite ultimately receiving 5.0/5 ratings from all 20 participants, points to a productive tension within this principle: the more a system preserves human judgment by withholding recommendations, the more it requires users to bring their own interpretive frame to the output. In regulated environments, this is not a design failure but a necessary feature. Future work should examine how to reduce the interpretive overhead of pattern outputs without collapsing that overhead into automated conclusions.

The Checklist as a Bridge

The checklist's immediate legibility relative to the pattern detection layer reflects a difference in the type of problem each addresses. The checklist solves a problem practitioners already recognize — inconsistent review preparation — and provides a tool that fits naturally into existing review practice. The pattern detection layer solves a problem prac-

titioners have not previously had the tools to name, and requires a conceptual reorientation before it can be used.

This suggests a practical adoption pathway: checklist-first deployment that establishes the dashboard's credibility within existing review practice, followed by progressive introduction of pattern detection outputs as users develop the interpretive frame to use them. The conceptual barrier to friction visibility is real but not insurmountable — all 20 participants endorsed the concept once demonstrated, and the final ratings were equivalent across both components.

Limitations

Several limitations bound the interpretation of these findings. The synthetic dataset underrepresents the terminological and structural variance of live review records, which may have reduced the discriminability of clustering outputs and contributed to the high acceptance ratings. Participants were recruited from a single team, limiting generalizability. The study does not establish effects on formal quality metrics — CAPA closure time, deviation recurrence rates, or audit findings — that would be required to demonstrate organizational impact.

6 Conclusion

This paper introduced review friction visibility as a design problem in regulated document environments. When deviation and CAPA review workflows generate delay, the coordination work producing that delay — revision loops, uneven reviewer burden, recurring comment patterns — remains invisible to any single participant. Responsibility is assigned retroactively, without access to the conditions that produced the outcome. Making that friction visible is a precondition for more equitable accountability, not merely a workflow efficiency measure.

The two-part prototype demonstrated that an external observability layer can surface hidden coordination work without modifying validated documentation systems or

automating quality decisions. All 20 participants rated both components 5.0/5, expressed adoption intent, and anticipated reduced accountability asymmetry from friction visibility. The study's central empirical finding is a dissociation between immediate legibility and final endorsement: the checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name. Both arrived at equivalent ratings. The interpretive overhead of the pattern detection layer is not a usability failure — it is a signal that making structural friction visible requires building a new problem category, not just a new interface.

Three contributions follow. First, review friction visibility is introduced as a design problem specific to regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. Second, the external metadata layer approach is demonstrated as a viable design strategy for surfacing coordination patterns within compliance boundaries. Third, the theoretical framing — coordination work as an upstream condition of accountability assignment — connects HCI concerns about workflow visibility to broader organizational questions about how responsibility is structured before formal judgment begins.

The most important direction for future work is deployment with live records in a production quality system. The checklist-first adoption pathway identified here provides a practical entry point; the progressive introduction of pattern detection outputs as practitioners develop the interpretive frame to use them is a research question as much as a deployment question. Both require engagement with the organizational conditions — compliance culture, role structure, accountability norms — that currently make review friction invisible.

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